

Module 10: Autonomous Agents

Plan, execute, verify, report

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Primary sources: Osmani, Chapter 10; Thomas & Hunt, Testing Strategies and Pragmatic Starter Kit.

Learning Outcomes

- Evaluate capabilities and limitations of autonomous agents.
 - Compare agent-based and traditional workflows.
 - Assess risks of increased autonomy.
 - Justify appropriate use of agents.
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From Copilot to Agent

In-IDE assistants operate under close human supervision.

Autonomous or background agents take a larger task and work through multiple steps.

This shifts the main question from “Can it write code?” to “Can we safely delegate this workflow?”

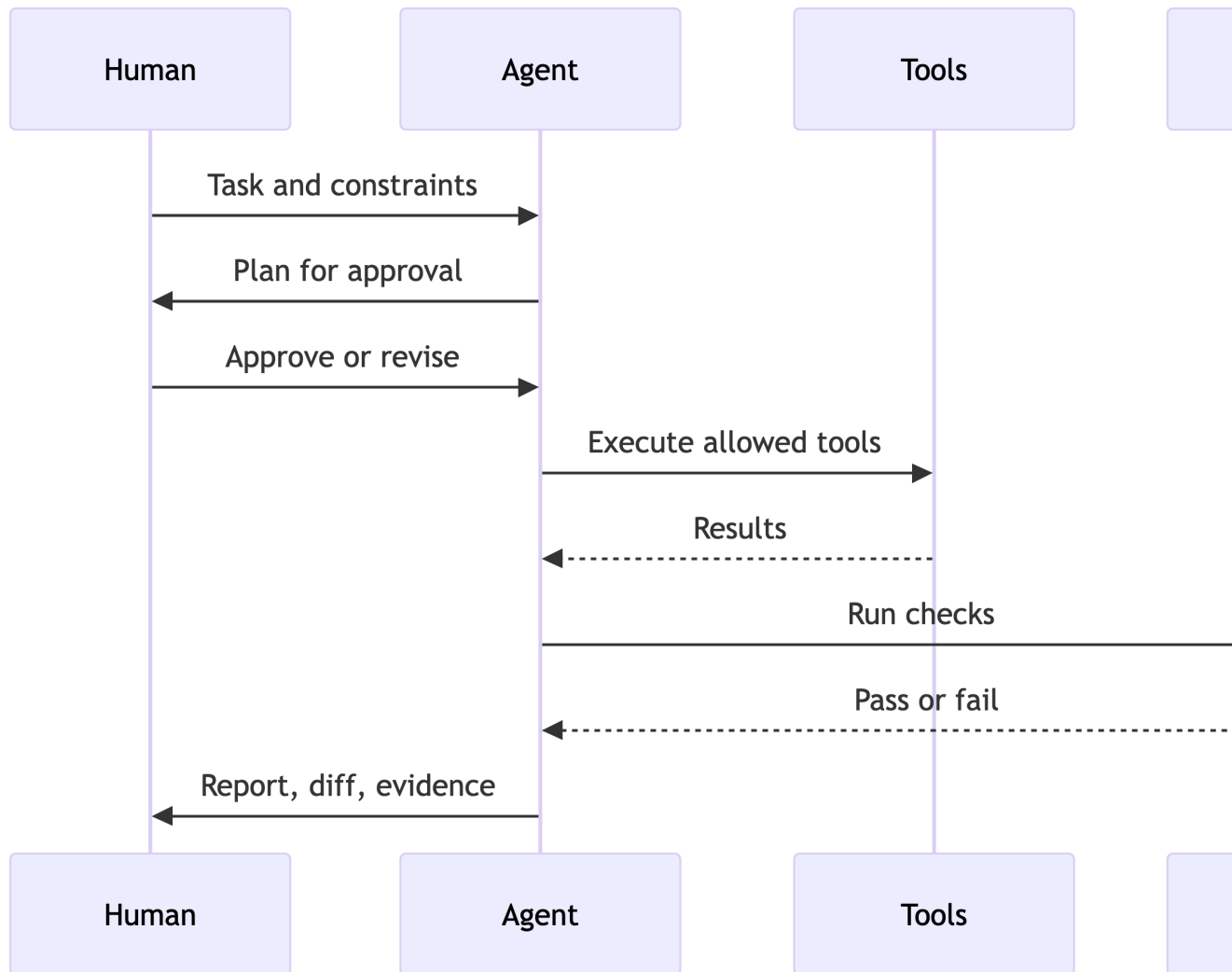
Agent Loop

Common autonomous workflow:

1. Plan.
2. Execute.
3. Verify.
4. Report.
5. Receive human feedback.

Your course agent should make this loop inspectable.

Agent Control Points



Appropriate Tasks

Good candidates:

- Small, well-scoped changes.
- Clear tests or checks.
- Low-risk refactors.

- Documentation updates.
- Repetitive maintenance.

Poor candidates:

- Ambiguous architecture decisions.
 - Security-critical changes without review.
 - Large rewrites.
 - Changes with weak verification.
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Autonomy Risks

- Wrong plan executed confidently.
 - Tool misuse.
 - Hidden side effects.
 - Incomplete verification.
 - Sunk-cost bias toward bad generated work.
 - Concurrent agent changes conflicting with humans.
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Human Oversight Checkpoints

Meaningful checkpoints happen before irreversible or high-risk action.

Examples:

- Approve plan before edits.
 - Approve file writes.
 - Review failing tests before retry loops.
 - Approve PR merge.
 - Require human signoff for sandbox exceptions.
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Testing Strategy for Agents

Agents need tests at multiple levels:

- Unit tests for tool dispatcher.
 - Integration tests for tool-result handling.
 - Golden examples for workflow output.
 - Safety tests for denied operations.
 - Manual review for architecture decisions.
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Lab 5 Final Connection

Your controlled workflow should demonstrate:

- Core Java agent loop.
 - Connected tool declarations and dispatcher.
 - Reproducible multi-step workflow.
 - Human oversight checkpoint.
 - Example run artifacts.
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Final Milestone 1 Connection

The proposal and architecture due this module should show how Labs 1-5 become a team project.

Your architecture should identify:

- Input.
 - Generation.
 - Evaluation.
 - Workflow.
 - Persistence.
 - Human oversight.
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Practice Activity

Evaluate a candidate task for delegation.

Classify it:

- Safe to automate.
- Safe with checkpoint.
- Human-only decision.

Explain the evidence needed for each classification.

Takeaways

- More autonomy requires stronger controls.
- Plan-execute-verify-report must be visible.
- Human review remains central to agentic workflows.